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Devices for analysis of a fluid

Abstract

An ultrasonic transducer is described, including a piezoelectric element, a fluid medium contact layer, a matching layer between the piezoelectric element and the fluid medium contact layer, and a backing layer. Ultrasound sensor devices utilising the ultrasonic transducer are also described, for use in systems for analysing a fluid such as milk.

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Background/Summary

STATEMENT OF CORRESPONDING APPLICATIONS

(1) This application is based on the provisional specification filed in relation to New Zealand Patent Application No. 756776, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to analysis of a fluid—more particularly ultrasound transducers for use in analysis of milk.

BACKGROUND

(3) The use of sensors to obtain information relating to milk collected from dairy animals is well known. Such information is used in decision making regarding such matters as processing of the milk, culling, breeding, medical treatment, animal specific feed rations as well as measurement of milk production efficiency.

(4) Numerous portable off-line analysers are known in the art for analysing a sample of milk to determine parameters such as fat, protein, lactose and total solids. Examples of such analysers using ultrasound analysis include the LactiCheck™ milk analyser by Page & Pedersen International, Ltd (www.pagepedersen.com); the Master milk analyser by Milkotester Ltd (www.milkotester.com); and the LACTOSCAN™ milk analyser by Milkotronic Ltd (www.lactoscan.com).

(5) Such off-line analysers are generally capable of relatively high precision measurements in comparison with commercially available in-line sensors. However, they have practical limitations associated with the collection of samples for analysis — requiring an operator to collect and deliver samples to the sensor.

(6) Sensors of other types are known for use in milking systems whereby samples are automatically extracted from the system for analysis. However, known ultrasound-based analysers are not well suited to this application, i.e. being fluidly connected to the milking system. For example, sensor having a measuring cell and one or more transducers needs to be suitable for exposure to milk as well as chemicals commonly used in cleaning milking systems. However, material selection for such an interface needs to be weighed against costs, and the ultrasonic transmission and acoustic impedance characteristics of the material.

(7) It is an object of the present invention to address the foregoing problems or at least to provide the public with a useful choice.

(8) Further aspects and advantages of the present invention will become apparent from the ensuing description which is given by way of example only.

SUMMARY

(9) The present disclosure provides ultrasound transducers, ultrasound sensor devices utilising said transducers, and systems for analysing a fluid sample.

(10) According to one aspect of the present disclosure, there is provided an ultrasonic transducer including a piezoelectric element, a fluid medium contact layer, a matching layer between the piezoelectric element and the fluid medium contact layer, and a backing layer.

(11) Arrangements and material properties of the elements of the transducer are provided for obtaining desired performance characteristics. Generally, it is envisaged that such arrangements may be used to reduce reflections of acoustic signals at material boundaries, or combinations of boundaries, with the objective of attaining a transducer performance for transmitting signals into, and receiving signals from, the sample with a near ideal damped characteristic as viewed in the time and frequency domains.

(12) More particularly, it is envisaged that the exemplary embodiments of the present disclosure may be used in sensing a characteristic of milk and/or water, where the nominal acoustic impedance of milk is 1.56 MRayl, while the nominal acoustic impedance of water is 1.49 MRayl. However, it should be appreciated that exemplary embodiments of the present disclosure may be used in the sensing of other fluids, particularly food or medical products. In examples, the fluid may be non-gaseous. In examples, the fluid may be a liquid. In examples the fluid may be a solution comprising solids suspended in a liquid, such as a slurry.

(13) It is envisaged that as the impedance of the fluid increases towards that of the fluid medium contact layer, the energy transmitted at that boundary may increase. The inverse is also to be

expected for lower impedance fluids, but while less transmitted energy and relatively more reflected energy can be expected, it is envisaged that a useful measurement may still be obtained. By way of example, ethanol (nominal acoustic impedance of 0.95 MRays) and honey (nominal acoustic impedance of 2.89 MRayls) are envisaged as being plausible measurement mediums for use with the transducer.

(14) In an exemplary embodiment the piezoelectric element may be one or more of: a lead metaniobate piezoelectric element; a piezoelectric element having an acoustic impedance of between 15 to 22 MRayls.

(15) Lead metaniobate is a commercially available piezoceramic material that has a relatively low acoustic impedance in comparison with other piezoceramic materials such as lead zirconate titanates (PZT). It is envisaged that this relatively low impedance may assist with reducing the impedance gap between the piezoelectric element, and the fluid medium contact layer. Further, this may assist with provision of a backing layer having a comparable acoustic impedance. Other piezoelectric materials are available (e.g. piezocomposites) that have relatively low impedance values and may be used in exemplary embodiments of the present disclosure, but it is envisaged that lead metaniobate may have particular application to embodiments in which limiting cost is a high priority consideration. In an exemplary embodiment, the lead metaniobate may be APC 3285 (available from APC International).

(16) In an exemplary embodiment the fluid medium contact layer may be one or more of: a polymer layer; a fluid medium contact layer having an acoustic impedance of between 2.5 to 3.5 MRayls.

(17) In an exemplary embodiment in which the fluid medium contact layer is a polymer layer, the polymer may be polysulfone. Polysulfone is generally characterised by properties well suited to the milking system environment such as chemical inertness, toughness, and thermal stability. Polysulfone has an acoustic impedance in the order of 2.78 MRayl, being relatively close to that of the fluid medium (e.g. milk or water) in comparison with other materials and thereby reducing reflections at the boundary between the contact layer and the fluid medium. Polysulfone also has suitable acoustic transmission properties in addition to the aforementioned properties, reducing attenuation in comparison with other materials. Polysulfone is also considered to be suitable for use in exemplary embodiments of the present disclosure in terms of price point, and ability to be machined or molded.

(18) It should be appreciated that reference to the polymer layer consisting of polysulfone is not intended to be limiting to all embodiments of the present disclosure. By way of example, it is envisaged that the polymer may be an amorphous polyamide (such as the Grilamid TR 90 product available from EMS Group)—particularly where intended for use in or with milking systems. It should also be appreciated that in exemplary embodiments the polymer layer may be a composite material, or a polymer with a non-polymer filler.

(19) In an exemplary embodiment, the thickness of the fluid medium contact layer may be configured to delay reflections of an acoustic signal from an interface between the fluid and the fluid medium contact layer by a predetermined number of wavelengths, or part wavelengths, of the acoustic signal. Reference to the thickness of the fluid medium contact layer should be understood to mean the dimension in the direction between the piezoelectric element and the fluid and contact layer interface. It is envisaged that there may be some mismatch of acoustic impedance at this interface as the result of other design constraints, resulting in reflections that may have an effect on signal detection and analysis. Delaying the reflections is expected to reduce their influence—i.e. received by the piezoelectric element after the majority of ringing from a pulse signal has died down.

(20) In an exemplary embodiment the number of wavelengths for the delay may be at least four. As such, the thickness of the fluid medium contact layer may be at least two wavelengths (such that the total path length is four wavelengths) of the acoustic signal—more particularly two

wavelengths at a low frequency cut-off of the design bandwidth.

(21) In another exemplary embodiment the thickness of the fluid medium contact layer may be about one quarter wavelength in length (e.g. between 0.2 to 0.3 wavelength) in order to provide a quarter wave matching effect as described below in relation to the matching layer.

(22) In an exemplary embodiment the matching layer may be one or more of: a hydrocarbon ceramic laminate layer; a matching layer having an acoustic impedance of between 4 to 10 MRayls; a circuit board layer having an acoustic impedance of between 4 to 10 MRayls.

(23) It should be appreciated that reference to a matching layer in the context of an ultrasound transducer is intended to mean an intermediate layer provided for reducing energy reflected between two materials in the acoustic path—more particularly, the piezoelectric element and the fluid medium contact layer.

(24) For this geometry with a harmonic plane wave at normal incidence, standard formulations exist for the transmission coefficient of acoustic energy from the piezoelectric element and the fluid contact layer (and vice-versa). These formulations show that when the thickness of the matching layer equates to $\frac{1}{4}$ wavelength within that material and $Z_{\text{sub}.0} = \sqrt{Z_{\text{sub}.1} \cdot Z_{\text{sub}.2}}$, where $Z_{\text{sub}.0}$ is the acoustic impedance of the matching layer with $Z_{\text{sub}.1}$ and $Z_{\text{sub}.2}$ being the acoustic impedances of the piezoelectric element and fluid contact layer, the transmission coefficient will be maximised and the frequency response approach that of an ideal damped transducer.

(25) It is envisaged that a matching layer impedance of between 4 to 10 MRayls may be suitable in exemplary embodiments of the present disclosure. In an exemplary embodiment the matching layer may have an acoustic impedance of between 5 to 8 MRayls.

(26) In exemplary embodiments in which the matching layer is a circuit board layer, it is envisaged that the matching layer may provide electrical connections to the piezoelectric element—i.e. in addition to its function in matching acoustic impedance between the piezoelectric element and the fluid medium contact layer.

(27) In an exemplary embodiment in which the matching layer is a hydrocarbon ceramic laminate, the hydrocarbon ceramic laminate may be a laminate such as the RO4000 series (available from Rogers Corporation), and more particularly RO4003C or an equivalent thereof.

(28) In addition to having a suitable acoustic impedance for use in exemplary embodiments of the present disclosure, being in the order of 6 MRayls, this material is fabricated with a copper layer (by way of electrodeposited copper foil) for PCB applications and has a construction more suitable to use in the present disclosure than other circuit board materials. A typical PCB fibreglass construction would tend to scatter the ultrasound signal, with associated losses. In contrast, the RO4000 material has a smaller grain size and thinner fibreglass reinforcing, with scattering being reduced as a result.

(29) In an exemplary embodiment, the thickness of the matching layer may provide quarter wave matching with the ultrasound signal. More particularly, the thickness of the matching layer may be between 20 to 30% of the signal wavelength at the centre frequency of the transducer (i.e. between 0.2 to 0.3 wavelengths of the acoustic signal). It should be appreciated that it may be generally desirable for the thickness of the matching material to have low variability in order to achieve this design parameter. In addition to the other properties described herein, the RO4000 material is considered to have a highly controlled thickness. For example, the RO4003 material used in exemplary embodiments of the present disclosure is available in a standard thickness of 0.203 mm, which approximates a quarter wavelength at a centre frequency of about 3.75 MHz.

(30) It should be appreciated that while the matching layer may be a single layer of material, in exemplary embodiments the matching layer may include two or more layers. However, control of the thickness of each of the layers becomes particularly important, and consequently the cost for implementing more than one matching layer can be unnecessarily high.

(31) It should be appreciated that alternative materials may be used for the matching layer in

exemplary embodiments—for example, an alumina composite.

(32) In an exemplary embodiment the backing layer may be one or more of: a backing layer having an acoustic impedance substantially that of the piezoelectric element; a backing layer having an acoustic impedance of between 15 to 20 MRayls; and a tungsten composite layer.

(33) The backing layer forms part of the acoustic path for ultrasound energy directed backwards from the piezoelectric element. It is considered desirable that the acoustic signal that propagates into the backing layer be rapidly absorbed—i.e. any acoustic signal propagating through the piezoelectric element, including that resulting from transmission excitation, does not reverberate for an extended period. More particularly, the ultrasound energy directed backwards from the piezoelectric element should essentially not return, being diminished in magnitude to substantially less than the forward going acoustic energy at that time (for example, by an order of magnitude).

(34) In an exemplary embodiment in which the backing layer is a tungsten composite layer, the tungsten composite may include tungsten particles of a first size, and tungsten particles of a second size. It is envisaged that this may assist with improving the ability of the backing layer to absorb the acoustic signal over a broader range of frequencies in comparison with a single particle size. Further, the relatively high density of tungsten is considered to aid in achieving a suitable acoustic impedance.

(35) In an exemplary embodiment, the larger particle may be a granulated tungsten powder—such as that made by crushing of sintered tungsten metal (for example GW-100270 available from Buffalo Tungsten Inc). In an exemplary embodiment, the smaller particle may be a fine tungsten powder (for example C20-491 available from Buffalo Tungsten Inc). It should be appreciated that the ratio of the two particle sizes may be adjusted according to the desired acoustic properties of the backing layer; in an exemplary embodiment the ratio may be in the order of 56:7 of the granulated powder to fine powder. It should be appreciated that the backing layer may include a suspension medium for the tungsten particles—for example, an epoxy resin.

(36) In an exemplary embodiment the backing layer may be manufactured using a centrifuge, such that there is a graduation in the density in the suspended particles due to the forces imparted by the centrifuge. It is believed that this graduation of density, and so a graduation in impedance, along the length of the backing layer may aid attenuation.

(37) According to one aspect of the present technology, an ultrasound transducer constructed in accordance with exemplary embodiments of the present disclosure may have one or more of: a centre frequency of between 1 to 10 MHz; a centre frequency of between 3 to 5 MHz; a centre frequency of between 3.5 to 4 MHz; and a centre frequency of about 3.75 MHz.

(38) Attenuation of ultrasound signals in a medium such as milk increases with frequency. As such, it is considered generally desirable to avoid a higher centre operating frequency, as this would require more energy to achieve a received signal of sufficient strength. It is also believed that higher frequency signals may also complicate the electronics design of associated circuitry of sensors utilizing the transducer(s), with associated issues in terms of cost and reliability. Higher frequencies may also require the design of the piezoelectric element and matching layer to be thinner, where complications in manufacture begin to arise due to the accuracy required, as well as becoming unsuitably fragile for assembly. Additionally, where adhesive is applied between layers, it is considered desirable for the thickness of the adhesive to be much less than a wavelength—at higher frequencies tighter tolerances are required for the surface finish of each of the mating surfaces to reduce discontinuities resulting from adhesive filling voids in the surfaces.

(39) Conversely, at lower design frequencies, the axial dimensions of key acoustic components will increase in large part in relation to the wavelength. Additionally, the attenuation of the acoustic signal propagating in the backing layer is significantly sensitive to frequency, as it is in part related to wave scattering. As a result, it is harder to attain backing performance at lower frequency. Generally speaking, a lower centre frequency will result in a lower resolution in time, and it is desirable to balance this effect against those associated with higher frequencies.

(40) For completeness, it should be appreciated that the centre frequency of the transducer is the cumulative result of the characteristics of the various layers (e.g. the piezoelectric element, fluid contact layer, matching layer and backing layer), and that the centre frequency may be adjusted accordingly by modification of these characteristics.

(41) According to one aspect of the present technology, an ultrasound transducer constructed in accordance with exemplary embodiments of the present disclosure may have one or more of: a -6 dB percentage bandwidth greater than 60%; a -6 dB percentage bandwidth of between 60% to 100%; a -6 dB bandwidth between 2 to 3 MHz.

(42) Bandwidth may be measured using a standard by which a narrow width pulsatile voltage is applied to the transducer to launch an ultrasonic wave. Either a second transducer may be used to detect this wave, or the transducer configured so the launched wave reflects and is detected by the same transducer. In both instances the received voltage is recorded and used to characterise the transducer performance.

(43) The ultrasound transducer requires sufficient bandwidth to achieve a desired resolution, more particularly temporal resolution of the acoustic signal propagation time through the fluid sample.

(44) Ultrasound transducers with a relatively narrow frequency response will produce a pulse containing several cycles, reducing the resolution. Conversely, a relatively wide frequency response provides a higher degree of damping which produces a shorter pulse in the time domain, resulting in higher resolution. However, performance in terms of resolution needs to be weighed against practical constraints such as costs and availability of piezoelectric elements having such characteristics. Manufacturing of such components requires more sophisticated processing techniques, with associated costs which are a significant barrier to their adoption in applications such as sensors for on-farm milk analysis. The technology of the present disclosure seeks to strike a balance between a desired level of performance in terms of resolution, and costs which might otherwise prohibit adoption of the technology.

(45) In an exemplary embodiment, the ultrasound transducer may include a housing. In an exemplary embodiment the housing may include a main body. In an exemplary embodiment, a portion of the main body may provide the fluid medium contact layer.

(46) In an exemplary embodiment the main body may include a projection through which the acoustic pathway of the transducer passes. It is envisaged that the projection may take the form of a solid cylindrical shaft, with a free end of the shaft providing a flat surface intended to face the fluid to be sensed in use.

(47) In exemplary embodiments the acoustic velocity in the material of the body, and therefore fluid medium contact layer, may be temperature dependent. It is contemplated that the temperature of the material may vary between measurements, and also vary more significantly that the fluid being sensed. The effect of this will depend on a number of factors, for example shaping of components of the transducer, insulation, temperature range experienced, and signal power input. However, it is envisaged that the shaft length, from the matching layer to the flat surface, may alter the extent to which ambient temperature effects sound speed measurements of the fluid. The fluid medium contact layer is exposed to ambient conditions, and also receives varying heat inputs from the piezoelectric element and other heating sources. This means that the temperature of the fluid medium contact layer may take longer to stabilise than the fluid, and may stabilise at different temperatures depending on the ambient conditions. Rather than control the temperature of the fluid medium contact layer, it is envisaged that this influence may be contained by reducing the length of the fluid medium contact layer as a percentage of the overall acoustic path length. In an exemplary embodiment in which two opposing transducers are used in an ultrasound sensor device, the length of the fluid medium contact layer may be less than 15% of the overall path length. Increasing the separation between transducer, or decreasing the length of the fluid medium contact layer, can further reduce the percentage if greater accuracy or wider ambient temperature ranges are required.

(48) In an exemplary embodiment the housing may include a cap, configured to be secured to the

main body. It is envisaged that the cap and main body may include complementary threads—but it should be appreciated that this is not intended to be limiting. For example, the cap and main body may be secured using one or more of: an interference fit, clips, fasteners, or any other suitable means known in the art.

(49) In an exemplary embodiment the ultrasound transducer may have a piezoelectric assembly including the piezoelectric element and the matching layer.

(50) In an exemplary embodiment the piezoelectric assembly may include an element holder. The element holder may include an aperture in which the piezoelectric element is located. In an exemplary embodiment the element holder may be made of a circuit board material.

(51) In an exemplary embodiment the matching layer may span the aperture of the element holder. In exemplary embodiments the matching layer may provide an electrical contact to the piezoelectric element—for example, where the matching layer is a circuit board.

(52) In an exemplary embodiment the piezoelectric assembly may include an electrical contact on the other side of the piezoelectric element from the matching layer. In an exemplary embodiment, the electrical contact may include a foil strip—for example a copper foil in the order of 35 μm or less in thickness. In an exemplary embodiment the foil may be arranged to span the aperture of the element holder and contact an entire face of the piezoelectric element. It is envisaged that the foil strip may be soldered to the element holder. It is envisaged that the foil thickness may be selected to provide a degree of robustness during assembly, but not be so thick so as to significantly influence the acoustic path to the backing layer.

(53) In an exemplary embodiment one or more electronic components may be provided on the element holder.

(54) In an exemplary embodiment, the housing body and/or piezoelectric assembly may be configured so as to locate the piezoelectric assembly in a desired position and orientation during assembly of the transducer. For example, the housing body may include a receiving portion configured to receive the piezoelectric assembly and shaped to restrict movement — particularly rotation or lateral movement.

(55) According to one aspect of the present disclosure there is provided an ultrasound sensor device, including: a hollow body configured to receive a fluid to be analysed, a first ultrasound transducer, and a second ultrasound transducer.

(56) In an exemplary embodiment the hollow body may be elongate. For example, it is envisaged that the hollow body may be a tube.

(57) In an exemplary embodiment the first and second ultrasound transducers may be arranged to face each other through the hollow body. In exemplary embodiments in which the hollow body is elongate the transducers may be disposed at distal ends of the hollow body.

(58) In an exemplary embodiment, the hollow body may be made of a metal. In an exemplary embodiment the hollow body may be made of a stainless steel.

(59) In ultrasound-based measuring cells used in off-line sensing of milk, the body is typically made of brass, which has a high heat transfer coefficient and can be constructed with a very thin wall, allowing it to quickly and precisely control milk temperature. However, brass is not resistant to the acidic chemicals commonly used in cleaning milking systems. The use of stainless steel may assist with providing resistance to such chemicals, enabling cleaning of the device using chemicals already in use within the wider system. This has further implications for assembly, as brass material allows the prior art ultrasonic transducer assemblies to be secured together, and to the measuring cell, in a relatively simple and robust way using solder. However, a solder-based assembly method is incompatible with materials such as stainless steel and polysulfone. As such, features of the present disclosure are intended to facilitate assembly of the transducer, and the ultrasound sensor device using the transducer, with one or more of the following considerations in mind: repeatability, secure and robust construction, achieving electrical connection to the piezoelectric element with insignificant impact on the acoustic performance, and doing so with the general

constraints of cost containment and material suitability for milk contact.

(60) According to one aspect of the present disclosure there is provided an ultrasound sensor device, including: a hollow body having an acoustically reflective surface and being configured to receive a fluid to be analysed, and a first ultrasound transducer facing the acoustically reflective surface.

(61) In exemplary embodiments the path length between piezoelectric elements of respective ultrasound transducers, or the total return path length between the piezoelectric element of an ultrasound transducer and the acoustically reflective surface, may be one of: greater than about 25 mm; between 25 mm to 100 mm; between 50 mm to 80 mm; greater than about 50 mm; between 60 mm to 75 mm; and about 70 mm. It is envisaged that such embodiments may be particularly applicable to embodiments in which the hollow body is tubular, and for use cases in which the fluid is milk.

(62) According to one aspect of the present disclosure there is provided a system for analysing a fluid, including: a ultrasound sensor device; a sample delivery device configured to deliver a sample of fluid from a fluid carrying and/or storing system to the ultrasound sensor device; and at least one processor configured to determine a characteristic of the sample of fluid based at least in part on a signal output from the ultrasound sensor device.

(63) In exemplary embodiments, the fluid may be milk extracted from a milking animal. It is envisaged that the present disclosure may have particular application to the analysis of milk during the transfer of milk from the point of extraction to a storage vessel. Milking systems typically include individual milk transport conduits from the points of extraction (for example, using a milking cluster including teat cups), joining to a common transport line for delivery to the storage vessel.

(64) In exemplary embodiments, the system may be embodied in a single unit—which may be referred to herein as a sensor. Various configurations of sensors, in terms of how the sensor is exposed to the fluid to be analysed, are known in the art. Terms such as “in-line”, “on-line”, “at-line”, “near-line” and “off-line” are used in the art to distinguish between these configurations—however there is a degree of inconsistency in their usage. For clarity, reference to an on-line sensor should be understood to mean a sensor which automatically extracts a sample of fluid from a fluid source (for example, a milk line or jar in the context of milking systems), and analyses the sample of fluid to determine at least one characteristic of the sample. As used herein, the term “on-line” may encompass embodiments in which the sample is returned to the source, or discarded. The terms “at-line” and “off-line” may be used in the art to distinguish between the environment in which the sensor is configured to operate. Both at-line and off-line sensors are configured to analyse a discrete sample of the fluid delivered to the sensor by an operator. At-line sensors (which may be referred to as “near-line” sensors) are generally intended to be located within the vicinity of the fluid source—for example, within a milking facility—while off-line sensors are primarily intended for use in a more environmentally controlled environment—for example, in a laboratory. In practice, particularly for milking operations, analysis of a sample by an off-line sensor may necessitate transport of the sample from the sample source to a remote facility. As used herein, the term “off-line” should be understood to refer to a sensor configuration in which a sample is collected from the fluid, and delivered to the sensor by an operator rather than an automated system. On-line and off-line sensors, as defined herein, may be distinguished from in-line sensors by the act of analysing a sample extracted from the fluid rather than analysing the flow itself. As such, on-line and off-line sensors may be referred to in the collective as “sample” sensors.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) Further aspects of the present invention will become apparent from the following description which is given by way of example only and with reference to the accompanying drawings in which:
- (2) FIG. 1 is a schematic diagram of an exemplary livestock management system in which an aspect of the present disclosure may be implemented;
- (3) FIG. 2 is a schematic diagram of an exemplary on-line sensor according to one aspect of the present disclosure;
- (4) FIG. 3A is a top view of an exemplary ultrasound sensor device according to one aspect of the present disclosure;
- (5) FIG. 3B is a side cross-section view of the ultrasound sensor device;
- (6) FIG. 4A is a perspective assembled view of an exemplary ultrasound transducer according to one aspect of the present disclosure;
- (7) FIG. 4B is a perspective exploded view of the ultrasound transducer;
- (8) FIG. 4C is a side cross-sectional view of the ultrasound transducer;
- (9) FIG. 5 is a perspective view of an exemplary piezoelectric assembly of the ultrasound transducer according to one aspect of the present disclosure, and
- (10) FIG. 6 is a graph of the frequency response of an exemplary ultrasound transducer according to one aspect of the present disclosure.

DETAILED DESCRIPTION

- (11) Exemplary embodiments are discussed herein in the context of analysis of milk. However, it should be appreciated that principles of the disclosure discussed herein may be applied to the analysis of other fluids.
- (12) FIG. 1 illustrates a livestock management system **100**, within which a local hardware platform **102** manages the collection and transmission of data relating to operation of a milking facility. The hardware platform **102** has a processor **104**, memory **106**, and other components typically present in such computing devices. In the exemplary embodiment illustrated the memory **106** stores information accessible by processor **104**, the information including instructions **108** that may be executed by the processor **104** and data **110** that may be retrieved, manipulated or stored by the processor **104**. The memory **106** may be of any suitable means known in the art, capable of storing information in a manner accessible by the processor **104**, including a computer-readable medium, or other medium that stores data that may be read with the aid of an electronic device. The processor **104** may be any suitable device known to a person skilled in the art. Although the processor **104** and memory **106** are illustrated as being within a single unit, it should be appreciated that this is not intended to be limiting, and that the functionality of each as herein described may be performed by multiple processors and memories, that may or may not be remote from each other. The instructions **108** may include any set of instructions suitable for execution by the processor **104**. For example, the instructions **108** may be stored as computer code on the computer-readable medium. The instructions may be stored in any suitable computer language or format. Data **110** may be retrieved, stored or modified by processor **104** in accordance with the instructions **108**. The data **110** may also be formatted in any suitable computer readable format. Again, while the data is illustrated as being contained at a single location, it should be appreciated that this is not intended to be limiting—the data may be stored in multiple memories or locations. The data **110** may also include a record **112** of control routines for aspects of the system **100**.
- (13) The hardware platform **102** may communicate with various devices associated with the milking facility, for example: in-line sensors **114a** to **114n** associated with individual milking clusters within the milking facility, and sample sensors in the form of on-line sensors **116a** to **116n** associated with the individual milking clusters or milk jars collecting milk from same.
- (14) Animal identification devices **118a** to **118n** are provided for determining an animal identification (“animal ID”) of individual animals entering, or within, the milking facility. More

particularly, the animal identification devices **118a** to **118n** may be used to associated an animal ID with each of the milking clusters associated with the in-line sensors **114a** to **114n** and on-line sensors **116a** to **116n**, such that the sensor data may be attributed to the individual animals. A variety of methodologies are known for the determination of an animal ID—for example a radio frequency identification (“RFID”) reader configured to read a RFID tag carried by the animal. In an alternative embodiment, or in conjunction with the animal identification devices **118a** to **118n**, a user may manually enter (or correct) animal IDs via a user device—examples of which are discussed below.

(15) The hardware platform **102** may also communicate with user devices, such as touchscreen **120** located within the milking facility for monitoring operation of the system, and a local workstation **122**. The hardware platform **102** may also communicate over a network **124** with one or more server devices **126** having associated memory **128** for the storage and processing of data collected by the local hardware platform **102**. It should be appreciated that the server **126** and memory **128** may take any suitable form known in the art—for example a “cloud-based” distributed server architecture. The network **124** potentially comprises various configurations and protocols including the Internet, intranets, virtual private networks, wide area networks, local networks, private networks using communication protocols proprietary to one or more companies—whether wired or wireless, or a combination thereof. It should be appreciated that the network **124** illustrated may include distinct networks and/or connections: for example a local network over which the user interface may be accessed within the vicinity of the milking facility, and an internet connection via which the cloud server is accessed. Information regarding operation of the system **100** may be communicated to user devices such as a smart phone **130** or a tablet computer **132** over the network **124**.

(16) Referring to FIG. 2, an exemplary sensor **200** is illustrated, which may be used (for example) as one or more of the on-line sensors **116a** to **116n**. In this exemplary embodiment, the on-line sensor **200** includes an ultrasound sensor device **300**, configured to perform ultrasound-based measurements of milk contained therein.

(17) The sensor **200** includes sample delivery device **202** configured to be connected to a source of the fluid to be sampled—for example milk tube **204** or milk jar **206**—and deliver a sample of the fluid to the ultrasound sensor device **300**.

(18) A controller **208** is provided to control the operation of the various components described, receive data obtained by the ultrasound sensor device **300**, and communicate over a network such as the network **124**.

(19) FIG. 3A and FIG. 3B illustrate an exemplary embodiment of the ultrasound sensor device **300**. The sensor device **300** includes a hollow body in the form of a stainless steel main tube **302**, having a first end **304a** and a second end **304b**. A first port tube **306a** is provided proximate the first end **304a**, while a second port tube **306b** is provided proximate the second end **304b**. In this exemplary embodiment, the port tubes **306** extend radially from the main tube **302** in opposing directions. In use, the port tubes **306** function as an inlet/outlet to and from the main tube **302**.

(20) Between the port tubes **306**, the exterior of the main tube **302** may be wrapped in coiled heating wires, for example an enamelled fine copper winding **308** around the main tube **302**, and a larger nichrome wire winding **310** coiled over the copper winding **308**. While not illustrated, it is envisaged that at least the copper windings **308** may also be provided on the port tubes **306**. The respective windings **308** and **310** may be connected in series, and current supplied to control temperature of the sensor device **300** and/or fluid being sensed. Further, the resistance of the windings **308** and/or **310** may be used to determine temperature. While not illustrated, it is also envisaged that insulating material may be provided over at least the main tube **302** in order to reduce the influence of ambient temperature and/or to increase efficiency when heating the fluid to measurement temperatures. It is envisaged that the insulating material may not cover the ends of the sensor device **300** to allow for heat dissipation from a first ultrasound transducer **400a** at the

first end **304a** of the main tube **302**, and a second ultrasound transducer **400b** at the second end **304b**.

(21) The first ultrasound transducer **400a** and the second transducer **400b** are arranged to face each other along the longitudinal axis of the main tube **302**. In use, one of the transducers **400** is configured as a transmitter, while the other is configured as a receiver.

(22) FIG. 4A to 4C illustrate an exemplary embodiment of the ultrasound transducer **400**. The transducer **400** includes a housing having a main body **402** and a cap **404** configured to be secured to the main body **402**. Referring to FIG. 4C, the main body **402** includes a first set of threads **406**, while the cap **404** includes a second set of threads **408** configured to engage the first set of threads **406**. The cap **404** includes a tool engaging portion, for example hexagonal head **410**. In this exemplary embodiment, the housing is made of polysulfone, as will be discussed further below.

(23) Referring to FIG. 4A, the main body **402** includes a slotted portion **412** configured to receive components of the transducer **400**. In the exemplary embodiment illustrated, a piezoelectric element holder (referred to herein as piezo holder **414**) is provided, having an aperture **416** configured to receive a piezoelectric element **418**. In this embodiment, the piezoelectric element **418** is a disk-shaped lead metaniobate piezoelectric element. The acoustic impedance of the piezoelectric element **418** is between 15 to 22 MRayls, more particularly in the order of 16 MRayls. In an exemplary embodiment the piezoelectric element is constructed of 0.5 mm thick APC3285. When heavily damped in the exemplary transducer design the resulting transducer has a centre frequency around 3.75 MHz.

(24) In this embodiment, the piezo holder **414** is made of a circuit board material for ease of forming electrical connections. The piezo holder **414** includes a rigid portion for locating the piezoelectric element **418** and an elongate flex portion **504** to act as an electrical connection. A piezoelectric contact member (referred to herein as piezo contact **420**) is provided beneath the piezo holder **414** to contact the piezoelectric element **418**. In this embodiment, the piezo contact **420** is made of RO4003C with a 35 μm electrodeposited copper foil layer, available from Rogers Corporation. In this exemplary embodiment, an electrical contact is provided on the opposing side of the piezo holder **414** (to the piezo contact **420**) in the form of a foil strip **422**—for example, a copper foil in the order of 33 μm in thickness.

(25) Above the foil strip **422**, a backing element **424** is provided. The backing element **424** is generally cylindrical in shape, having locating wings **426a** and **426b** on opposing sides. The locating wings **426** align with the slotted portion **412** of the main body **402**, to assist with maintaining the position of the backing element **424** during assembly. A PTFE gasket **428** is provided between the backing element **424** and the cap **404**.

(26) Referring to FIG. 4C, below the piezo contact **420** the main body **402** includes a cylindrical shaft **430**. The distal end of the shaft **430** has a flat surface **432**, which in use is presented to the interior of the main tube **302** (as shown in FIG. 3B). The thickness of the main body **402** from the flat surface **432** to the piezo contact **420** is influenced by several factors. Firstly, the temperature of polysulfone has an effect on acoustic performance (more particularly sound speed), and the main body **402** is subject to heat inputs from the piezoelectric element **418** and windings **308** and **310**, as well as fluctuations in the ambient temperature. Secondly, reflections of the acoustic signal will occur at the boundary between the flat surface **432** and the fluid, which if not accounted for will interfere with signal analysis. As such, there is a balance to be struck between reducing the thickness of the polysulfone as a percentage of the overall signal transmission path, and achieving a sufficient thickness so as to delay the reflections.

(27) Further, there are practical considerations for assembly of the transducer **400**—for example, the shaft **430** being of a sufficient length for fitting to the main tube **302** of the sensor device **300**. It should also be appreciated that the diameter of the shaft **430** may be sized relative to the internal diameter of the main tube **302** to produce a seal, as well as mechanically securing the transducer **400** to the main tube **320**.

(28) In the exemplary embodiment illustrated in FIG. 3A and 3B, the respective piezoelectric elements of the transducers **400a** and **400b** are spaced apart at 69 mm (comprising a 59 mm gap between end surfaces, and a polysulfone thickness of about 5 mm). The thickness of the polysulfone of each transducer **400** is about 5 mm, which in combination represents just under 15% of the total path length. It is also considered desirable for reflections to be delayed by at least four wavelengths—i.e. requiring the thickness of the polysulfone to be at least two wavelengths. In a design having a 2.5 MHz low frequency cut-off, and 2240 m/s nominal sound speed in polysulfone, the wavelength of the signal through the main body **402** is 0.896 mm. A 5 mm thickness provides a total path length of eleven wavelengths, which satisfies this design criteria.

(29) With the acoustic impedance of the piezoelectric element being in the order of 16 MRayls, it is desirable for the acoustic impedance of the backing element **424** to be substantially the same. In the exemplary embodiment described herein, the backing element **424** is made of a tungsten composite, including tungsten particles of a relatively large size (more particularly granulated tungsten powder—such as GW-100270 available from Buffalo Tungsten Inc), and tungsten particles of a relatively small size (more particularly fine tungsten powder—such as C20-491 available from Buffalo Tungsten Inc). It is envisaged that this may assist with improving the ability of the backing element to absorb the acoustic signal over a broader range of frequencies in comparison with a single particle size. Further, the relatively high density of tungsten is considered to aid in achieving a suitable acoustic impedance. In this exemplary embodiment, the tungsten particles are suspended in an epoxy resin, for example EpoTek 301 available from Epoxy Technology Inc. In an exemplary embodiment, the ratio of GW-100270:C20-491:EpoTek 301A may be in the order of 56:7:3.5 (with the second part of the epoxy EpoTek 301B later added at 0.875). It is envisaged that the unset mixture may be spun in a centrifuge so as to promote settling of the tungsten particles towards an end of the backing element **424** which is proximate the piezoelectric element **418** in use.

(30) The polysulfone of the main body **402** may be Sustason PSU rod stock produced from non UV-stabilised polysulphone resin (available from Rochling Sustaplast SE & Co. KG), which has a nominal acoustic impedance in the order of 2.78 MRayl. Due to the differential in acoustic impedance between the polysulfone and the piezoelectric material it is desirable to include an impedance matching layer. The piezo contact **420** is configured for this purpose. An ideal matching layer should have an acoustic impedance ($Z_{\text{sub.0}}$) that is the square root of the product of the acoustic impedance of the adjacent materials (i.e. $Z_{\text{sub.1}}$ and $Z_{\text{sub.2}}$)—i.e.

$Z_{\text{sub.0}} = \sqrt{(Z_{\text{sub.1}} \cdot Z_{\text{sub.2}})}$. For the lead metaniobate piezoelectric material and polysulfone interface of the present example, the ideal acoustic impedance of the piezo contact **420** would be 6.67 MRayls. While it is generally preferable for the acoustic impedance of the matching layer to be as close as possible to the ideal value, in practice the impedance may be within a wider range and still produce a useful result—particularly where other design constraints are present.

(31) The thickness of the piezo contact **420** is also designed to provide quarter wave matching with the ultrasound signal. More particularly, the thickness of the piezo contact **420** may be between 20 to 30% of the signal wavelength at the centre operating frequency of the transducer **400** (i.e. 3.75 MHz). As such, with a nominal acoustic impedance of about 6 MRayls and thickness of about 0.203 mm, the RO4003C piezo contact **420** is considered suitable for impedance matching. Further, the RO4003C material can be supplied with an electrodeposited copper foil, which is used in the exemplary embodiment to provide electrical contact between the piezo holder **414** and the bottom face of the piezoelectric element **418**. In the exemplary embodiment the electrodeposited foil is 35 μm thick.

(32) FIG. 5 illustrates a piezoelectric assembly **500**, including the piezo holder **414**, piezoelectric element **418**, piezo contact **420**, and foil strip **422** as previously described. In an exemplary embodiment, prior to assembly of the transducer **400**, the piezo contact **420** may be soldered to the piezo holder **414**, and one end of the foil strip **422** soldered to the piezo holder **414**. One or more

electronic components **502** may also be soldered to the piezo holder **414**. It may be seen that the piezo holder **414** also includes a flex portion **504**—which may be used to provide electric connections to associated circuitry. The resulting sub-assembly is used to assist in assembly of the transducer **400**.

(33) During assembly, the piezoelectric element **418** is inserted into the aperture **416** of the piezo holder **414**, beneath the foil strip **422**. The thickness of the piezo holder **414** is such that the piezoelectric element **418** projects above it, to ensure contact with the foil **422** (and therefore backing element **424**). The piezoelectric assembly **500** is then inserted into the slotted portion **412** of the main body **402**. The slotted portion **412** and piezoelectric assembly **500** are shaped such that the piezoelectric element **418** is centred over the shaft **430** of the main body **402**.

(34) During assembly, an epoxy adhesive (for example, Scotch-Weld™ Epoxy Adhesive EC-2216 B/A available from 3M Company) is used between the piezoelectric element **418**, the piezo contact **420** and the foil strip **422** for good acoustic contact, and between the piezo contact **420** and the main body **402**, and between the foil strip **422** and the backing element **424**.

(35) The backing element **424** is inserted into the slotted portion **412**, contacting the top of the foil strip **422**. The cap **404** is then screwed on to the main body **402** until a specified torque is achieved, to squeeze the epoxy from between the various layers and provide a desired pressure as the epoxy sets. It is envisaged that the residual epoxy may be sufficiently thin so as to have a negligible effect on acoustic transmission properties or electrical contact. During assembly a fillet of epoxy is also applied to cover the electronic component **502** and portions of the piezo holder **414** adjacent the housing, to seal and encapsulate the transducer components.

(36) FIG. **6** illustrates the frequency response **600** of an exemplary ultrasound transducer **400** constructed in accordance with the description above. The ultrasound transducer **600** has a centre frequency **602** of about 3.75 MHz, with a lower -6 dB frequency limit **604** of about 2.5 MHz and an upper -6 dB frequency limit **606** of about 4.9 MHz. This provides a -6 dB bandwidth of about 2.4 MHz, which may be expressed as a -6 dB percentage bandwidth of about 64%. For completeness, it will be appreciated that variation in the tested bandwidth may be expected between different batches of the transducer, and between individual examples of the transducer design.

(37) All references, including any patents or patent applications cited in this specification are hereby incorporated by reference. No admission is made that any reference constitutes prior art. The discussion of the references states what their authors assert, and the applicants reserve the right to challenge the accuracy and pertinency of the cited documents. Reference to any prior publications in this specification is not, and should not be taken as, an acknowledgement or any form of suggestion that that prior art forms part of the common general knowledge in the field of endeavour in any country in the world.

(38) Unless the context clearly requires otherwise, throughout the description and the claims, the words “comprise”, “comprising”, and the like, are to be construed in an inclusive sense as opposed to an exclusive or exhaustive sense, that is to say, in the sense of “including, but not limited to”. The entire disclosures of all applications, patents and publications cited above and below, if any, are herein incorporated by reference.

(39) The invention may also be said broadly to consist in the parts, elements and features referred to or indicated in the specification of the application, individually or collectively, in any or all combinations of two or more of said parts, elements or features.

(40) Where in the foregoing description reference has been made to integers or components having known equivalents thereof, those integers are herein incorporated as if individually set forth.

(41) It should be noted that various changes and modifications to the presently preferred embodiments described herein will be apparent to those skilled in the art. Such changes and modifications may be made without departing from the spirit and scope of the invention and without diminishing its attendant advantages. It is therefore intended that such changes and

modifications be included within the present invention.

(42) Aspects of the present invention have been described by way of example only and it should be appreciated that modifications and additions may be made thereto without departing from the scope thereof.

Claims

1. An ultrasonic transducer including: a piezoelectric element, wherein the piezoelectric element has an acoustic impedance of between 15 to 22 MRayls; a fluid medium contact layer, wherein the fluid medium contact layer is a polymer layer; a matching layer between the piezoelectric element and the fluid medium contact layer, wherein the matching layer has an acoustic impedance of between 4 to 10 MRayls; and a backing layer, wherein the backing layer has an acoustic impedance of between 15 to 20 MRayls.
2. The ultrasonic transducer of claim 1, wherein the piezoelectric element is a lead metaniobate piezoelectric element.
3. The ultrasonic transducer of claim 1, wherein the polymer layer is polysulfone.
4. The ultrasonic transducer of claim 1, wherein the thickness of the fluid medium contact layer is configured to delay reflections of an acoustic signal from an interface between the fluid medium contact layer and a fluid in contact with the fluid medium contact layer by a predetermined number of wavelengths, or part wavelengths, of the acoustic signal.
5. The ultrasonic transducer of claim 4, wherein the thickness of the fluid medium contact layer is greater than two wavelengths of the acoustic signal.
6. The ultrasonic transducer of claim 1, wherein the matching layer is a hydrocarbon ceramic laminate layer.
7. The ultrasonic transducer of claim 1, wherein the matching layer has an acoustic impedance of between 5 to 8 MRayls.
8. The ultrasonic transducer of claim 1, wherein the thickness of the matching layer is configured to provide quarter wave matching with the acoustic signal.
9. The ultrasonic transducer of claim 8, wherein the thickness of the matching layer is between 0.2 to 0.3 wavelengths of the acoustic signal at the centre frequency of the ultrasonic transducer.
10. The ultrasonic transducer of claim 1, wherein the matching layer is a circuit board layer.
11. The ultrasonic transducer of claim 10, wherein the circuit board layer provides electrical connections to the piezoelectric element.
12. The ultrasonic transducer of claim 1, wherein the backing layer has an acoustic impedance substantially that of the piezoelectric element.
13. The ultrasonic transducer of claim 1, wherein at least one of the following is true: the backing layer is a tungsten composite layer; the tungsten composite of the tungsten composite layer includes first tungsten particles of a first size, and second tungsten particles of a second size; the first tungsten particles are a granulated tungsten powder, and the second tungsten particles are a fine tungsten powder; a ratio of granulated tungsten powder to fine tungsten powder in the tungsten composite is about 56:7; the backing layer comprises a graduation in density of the first tungsten particles and the second tungsten particles; and any combination thereof.
14. The ultrasonic transducer of claim 1, wherein the ultrasonic transducer has a centre frequency of one of: between 1 to 10 MHz, between 3 to 5 MHz, 3.5 to 4 MHz, and about 3.75 MHz.
15. The ultrasonic transducer of claim 1, wherein the ultrasonic transducer has a -6 dB percentage bandwidth greater than 60%.
16. The ultrasonic transducer of claim 1, wherein the -6 dB bandwidth is between 2 to 3 MHz.
17. The ultrasonic transducer of claim 1, wherein at least one of the following is true: the ultrasonic transducer further includes a housing having a main body, wherein a portion of the main body provides the fluid medium contact layer; the main body includes a projection through which an

acoustic pathway of the ultrasonic transducer passes; the ultrasonic transducer further includes a piezoelectric assembly, the piezoelectric assembly including the piezoelectric element and the matching layer; the piezoelectric assembly includes an element holder having an aperture in which the piezoelectric element is located; wherein the element holder is made of a circuit board material; the matching layer spans the aperture of the element holder; the piezoelectric assembly includes an electrical contact on an opposing side of the piezoelectric element from the matching layer; the main body includes a slotted portion configured to receive the piezoelectric assembly, wherein the ultrasonic transducer further includes a cap configured to be secured to the main body to hold the piezoelectric assembly in place within the slotted portion; and any combination thereof.

18. A system for analysing a fluid, including: a ultrasound sensor device comprising an ultrasonic transducer as claimed in claim 1, a sample delivery device configured to deliver a sample of fluid from a fluid carrying and/or storing system to the ultrasound sensor device; and at least one processor configured to determine a characteristic of the sample of fluid based at least in part on a signal output from the ultrasound sensor device.

19. The system of claim 18, wherein the fluid is one of: a liquid, a solution including a liquid, and milk.

20. An ultrasound sensor device, including: an elongate hollow body configured to receive a fluid to be analysed; a first ultrasound transducer, provided at a first end of the hollow body, wherein the first ultrasound transducer includes: a piezoelectric element, wherein the piezoelectric element has an acoustic impedance of between 15 to 22 MRayls; a fluid medium contact layer, wherein the fluid medium contact layer is a polymer layer; a matching layer between the piezoelectric element and the fluid medium contact layer, wherein the matching layer has an acoustic impedance of between 4 to 10 MRayls; and a backing layer, wherein the backing layer has an acoustic impedance of between 15 to 20 MRayls.

21. The ultrasound sensor device of claim 20, further including a second ultrasound transducer of the same configuration as the first ultrasound transducer, wherein the second ultrasound transducer is provided at a second end of the hollow body and facing the first ultrasound transducer.

22. The ultrasound sensor device of claim 21, wherein a length of the respective fluid medium contact layers of the first ultrasound transducer and the second ultrasound transducer is less than 15% of the overall path length.

23. The ultrasound sensor device of claim 21, wherein a path length between the respective piezoelectric elements of the first ultrasound transducer and the second ultrasound transducer is one of: greater than about 25 mm, between 25 mm to 100 mm, and about 70 mm.

24. The ultrasound sensor device of claim 20, wherein the hollow body is made of a metal.

25. The ultrasound sensor device of claim 20, wherein the hollow body comprises an acoustically reflective surface, wherein the first ultrasound transducer faces the acoustically reflective surface.

26. The ultrasound sensor device of claim 25, wherein a total return path length between the piezoelectric element of the first ultrasound transducer and the acoustically reflective surface is one of: greater than about 25 mm, between 25 mm to 100 mm, and about 70 mm.
